

7. Encuentre el o los números $z \in \mathbb{C}$ que satisfacen simultáneamente las siguientes condiciones:

$$\begin{cases} |z| = 5 \\ \arg(3-z) = \frac{-\pi}{2} \end{cases}$$

$$z = a + bi$$

$R/z = 3 + 4i$

$$\begin{aligned} & b > 0 \\ & + \frac{\pi}{2} \rightarrow a = 0 \\ & b < 0 \end{aligned}$$

$$\arg(3-z) = \frac{-\pi}{2}$$

$$3 - (a + bi)$$

$$3 - a - bi$$

$$a = 0$$

$$(3-a) + (-b)i$$

$$b < 0$$

$$a$$

$$b$$

$$3 - a = 0$$

$$-b < 0$$

$$a = 3$$

$$b > 0$$

$$|z| = 5$$

$$|a + bi| = 5$$

$$|(a) + (b)i| = 5$$

$$a^2 + b^2 = 5^2$$

$$a = 3$$

$$3^2 + b^2 = 25$$

$$b^2 = 16$$

$$b = \pm 4$$

$$b > 0$$

$$b = 4$$

$$\boxed{R/ 3 + 4i}$$

14. Encuentre el o los números $z \in \mathbb{C}$ que satisfacen simultáneamente las siguientes condiciones:

$$\operatorname{Re} z = -3\sqrt{2}i$$

$$\begin{aligned} z &= a + bi \\ \bar{z} &= a - bi \end{aligned} \quad \begin{cases} |z - \bar{z}| = 6\sqrt{2} \\ \arg(\bar{z} - 3\sqrt{2}) = \frac{3\pi}{4} \end{cases}$$

$$\arg(\bar{z} - 3\sqrt{2}) = \frac{3\pi}{4} \rightarrow \neq \pm \frac{\pi}{2}$$

$$a - bi - 3\sqrt{2}$$

$$\frac{b}{a} = \tan(\theta)$$

$$\underbrace{(a - 3\sqrt{2})}_a + \underbrace{(-b)}_b i = \frac{3\pi}{4}$$

$$\frac{-b}{a - 3\sqrt{2}} = \tan\left(\frac{3\pi}{4}\right) \quad \frac{3\pi}{4} \rightarrow 135^\circ, \frac{3\pi}{4}, \frac{180}{\pi}$$

$$\frac{-b}{a - 3\sqrt{2}} = -1$$

$$135^\circ \quad a < 0 \quad b > 0$$

$$90 < 135 < 180$$

$$-b = 3\sqrt{2} - a$$

$$b = a - 3\sqrt{2} \quad \left. \begin{array}{l} b = a - 3\sqrt{2} \\ a = b + 3\sqrt{2} \end{array} \right\} \text{Cualquiera}$$

$$a = b + 3\sqrt{2}$$

Cuadrante	a	b
1) $0 < \theta < \frac{\pi}{2}$ $0 < a < 90$	> 0	> 0

2) $\frac{\pi}{2} < \theta < \pi$ $90 < a < 180$	< 0	> 0
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3) $\pi < \theta < \frac{3\pi}{2}$ $180 < a < 270$	< 0	< 0
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4) $\frac{3\pi}{2} < \theta < 2\pi$ $270 < a < 360$	> 0	< 0
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En cualquier caso que el arg NO sea $\pm \pi$ v $\pm \frac{\pi}{2}$ para sacar el ángulo se $\theta = \left(\frac{b}{a} = \tan(\theta)\right)$

$$|z - \bar{z}| = 6\sqrt{2}$$

$$|a+bi - (a-bi)| = 6\sqrt{2}$$

$$|\cancel{a}+bi - \cancel{a}+bi| = 6\sqrt{2}$$

$$|(2b)i| = 6\sqrt{2}$$

$$4b^2 = 72$$

$$b^2 = 18$$

$$b = \pm 3\sqrt{2}$$

$$b > 0$$

$$b = 3\sqrt{2}$$

$$\boxed{z = -3\sqrt{2}}$$

P2-82-2025

#1 Sea z el número complejo tal que $z = \overline{(2+3i)} \div (1-i) - (3-i) \cdot i^{2025}$. Determine $\text{Im}(z)$.
Muestre el procedimiento paso a paso en cada operación realizada. (4 Pts)

$$\overline{z} = \overline{2+3i} \rightarrow z \rightarrow = 2-3i$$

$$\boxed{\frac{2-3i}{1-i}} - \underbrace{(3-i)}_{22} \cdot \underbrace{i^{2025}}_{23}$$

21

$$\text{21} = \frac{2-3i}{1-i} \cdot \frac{1+i}{1+i}$$

$$= \frac{(2-3i)(1+i)}{1^2 - i^2}$$

$1 - -1 = 2$

$$= \frac{2+2i-3i-3i^2}{2}$$

$$= \frac{2+2i-3i+3}{2} = \frac{5-i}{2}$$

$$\text{23} = i^{2025} \xrightarrow{n-7 \cdot \text{int}(\frac{n}{7})} i^{2025-7 \cdot \text{int}(506,25)}$$

$2025 - 7 \cdot 506$
 1

$$i^{2025} = i^1 = i$$

$$\begin{aligned} (3-i) \cdot i^{2025} &= (3-i) \cdot i \\ &= 3i - i^2 \\ &= 3i + 1 \\ &= 1 + 3i \end{aligned}$$

$$\frac{2-3i}{1-i} - \frac{(3-i)}{2} \cdot i^{2025}$$

22

22

23

$$\frac{5-i}{2} - (1+3i)$$

$$\frac{5}{2} - \frac{i}{2} - 1 - 3i$$

$$\text{Im} \left(\frac{3}{2} - \frac{7}{2}i \right)$$

$$\text{Im} = \frac{-7}{2}$$

#2 Considere la siguiente ecuación

$$x^3 + (-1 + 3i)x^2 + (9 - 4i)x + (-5 - 5i) = 0.$$

Si se sabe que $1 + i$ es una solución de la ecuación, determine en $\mathbb{C}$ su conjunto solución. Muestre de manera detallada los procedimientos que le permitan obtener las demás soluciones de la ecuación.

(4 Pts)

1	$-1+3i$	$9-4i$	$-5-5i$	
	$1+i$	$4i-4$	$5+5i$	$1+i$
1	$4i$	5	0	$(x - (1+i))$

$$x^2 + (4i)x + 5 = -5i - i$$

$$\begin{array}{rcl} x & \times & 5i = 5i \\ x & \times & = i = \frac{-i}{4i} \end{array}$$

$$\begin{array}{l} 4i(1+i) \\ 4i + 4i^2 \\ 4i - 4 \end{array}$$

$$(x + 5i)(x - i)(x - (1+i))$$

$$x + 5i = 0 \quad x - i = 0 \quad x - (1+i) = 0$$

$$x = -5i \quad x = i \quad x = 1+i$$

$$S = \{-5i, i, 1+i\}$$

#3 Sea z un número complejo ubicado en el **segundo cuadrante** del plano complejo que satisfice

$$z^3 = -2 + 2i\sqrt{3}.$$

Calcule explícitamente z y determine las coordenadas polares de z^2 .

(4 Pts)

$$z^n = r \operatorname{cis}(\theta) \quad r = \sqrt{a^2 + b^2}$$

$$z = \sqrt[n]{r} \cdot \operatorname{cis}\left(\frac{\theta + 2\pi k}{n}\right) \quad \theta = \arctan\left(\frac{b}{a}\right)$$

$$k = 0, \dots, n-1$$

$$z^3 = -2 + 2\sqrt{3}i$$

$$-2 + 2\sqrt{3}i \quad a = -2 \quad b = 2\sqrt{3}$$

a negativo $\theta \rightarrow +\pi$
 b positivo

$$r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = 4$$

$$\theta = \arctan\left(\frac{2\sqrt{3}}{-2}\right) + \pi = \frac{2\pi}{3}$$

$$z = 4 \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$n=3$$

$$z^3 = -2 + 2\sqrt{3}i$$

$$z^n = r \operatorname{cis}(\theta)$$

$$z = \sqrt[n]{r} \cdot \operatorname{cis}\left(\frac{\theta + 2\pi k}{n}\right)$$

$$z^3 = 4 \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$k = 0, \dots, n-1$$

$$z = \sqrt[3]{4} \operatorname{cis}\left(\frac{\frac{2\pi}{3} + 2\pi k}{3}\right)$$

$$n=3$$

$$k = 0, 1, 2$$



$$n-1$$

$$3-1$$

$$2$$

$$k=0, \dots, n-1$$

$$n=3$$

$$k=0, 1, 2$$

$$z = \sqrt[3]{4} \cdot \text{cis} \left(\frac{\frac{2\pi}{3} + 2\pi k}{3} \right)$$

↑

$$k-1$$

$$3-1$$

$$2$$

$$k=0 \rightarrow \sqrt[3]{4} \cdot \text{cis} \left(\frac{\frac{2\pi}{3} + 2\pi \cdot 0}{3} \right)$$

$$\sqrt[3]{4} \cdot \text{cis} \left(\frac{2\pi}{9} \right)$$

$$\frac{2\pi}{9}, \frac{360}{9} = 40$$

$$0 < 40 < 90$$

1 Quadrante X

$$k=1 \rightarrow \sqrt[3]{4} \cdot \text{cis} \left(\frac{\frac{2\pi}{3} + 2\pi \cdot 1}{3} \right)$$

$$\sqrt[3]{4} \cdot \text{cis} \left(\frac{8\pi}{9} \right)$$

$$\frac{8\pi}{9}, \frac{360}{9} = 160$$

$$90 < 160 < 180$$

2 Quadrante ✓

$$k=2 \rightarrow \sqrt[3]{4} \cdot \text{cis} \left(\frac{\frac{2\pi}{3} + 2\pi \cdot 2}{3} \right)$$

$$\sqrt[3]{4} \cdot \text{cis} \left(\frac{14\pi}{9} \right)$$

$$\frac{14\pi}{9}, \frac{360}{9} = 280$$

$$270 < 280 < 360$$

4 Quadrante X

$$z = \sqrt[3]{4} \operatorname{cis} \left(\frac{8\pi}{9} \right)$$

$$z^2 = ?$$

$$z^2 = \left(\sqrt[3]{4} \operatorname{cis} \left(\frac{8\pi}{9} \right) \right)^2 \quad \begin{matrix} [r \operatorname{cis}(\theta)]^n \\ r^n \cdot \operatorname{cis}(\theta \cdot n) \end{matrix}$$

$$(\sqrt[3]{4})^2 \cdot \operatorname{cis} \left(\frac{8\pi \cdot 2}{9} \right)$$

$$\sqrt[3]{16} \cdot \operatorname{cis} \left(\frac{16\pi}{9} \right)$$

$(x, y) \rightarrow \text{Reales}$

$(a, b) \rightarrow \text{Rectangular}$

$(r, \theta) \rightarrow \text{Polares}$

$$\boxed{\left(\sqrt[3]{16}, \frac{16\pi}{9} \right)}$$

#4 Sea $z = a + bi$ un número complejo que cumple las siguientes condiciones:

$$\begin{cases} |z + 3 - 3i| = \sqrt{74} \\ \text{Arg}(z + 4i) = \pi \end{cases} \rightarrow b=0 \begin{cases} a > 0 \\ a < 0 \end{cases}$$

Determine todos los números z que cumplan, de manera simultánea, ambas condiciones.

$$\text{Arg}(z + 4i) = \pi$$

$$a + bi + 4i$$

$$(a) + (b + 4)i$$

$$z + 4i$$

$$b = 0 \quad a < 0$$

$$z - 4i$$

$$b + 4 = 0 \quad a < 0$$

$$b = -4$$

$$b = 0 \quad a > 0$$

$$|z + 3 - 3i| = \sqrt{74}$$

$$|a + bi + 3 - 3i| = \sqrt{74}$$

$$|(a + 3) + (b - 3)i| = \sqrt{74}$$

$$(a + 3)^2 + (b - 3)^2 = 74$$

$$a^2 + 6a + 9 + 79 = 74$$

$$a^2 + 6a + 58 - 79 = 0$$

$$a^2 + 6a - 21 = 0$$

$$a \quad 8 = 8a$$

$$a \quad -2 = -2a$$

$$6a$$

$$(a + 8)(a - 2) = 0$$

$$a = -8$$

$$a = 2$$

$$a < 0$$

$$a = -8$$

$$b = -4$$

$$z = -8 - 4i$$

#5 En un laboratorio de ingeniería se modelan dos procesos con las matrices:

$$A = \begin{pmatrix} 2 & -1 & \alpha \\ 0 & 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 0 \\ 4 & -6 & -2 \end{pmatrix},$$

donde $\alpha \in \mathbb{R}$. El resultado combinado de ambos procesos se representa por la matriz:

$$C = A + B.$$

Se sabe que al multiplicar C por la matriz $Z = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, se obtiene $C \cdot Z = \begin{pmatrix} 8 \\ \beta \end{pmatrix}$, con $\beta \in \mathbb{R}$.

Determine los valores de α y β .

(3 Pts)

$$C = \begin{pmatrix} 2 & -1 & x \\ 0 & 3 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 0 \\ 4 & -6 & -2 \end{pmatrix}$$

$$C = \begin{pmatrix} 3 & 1 & x \\ 4 & -3 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 & x \\ 4 & -3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ y \end{pmatrix}$$

$$\begin{pmatrix} 3 - 1 + 2x \\ 4 + 3 - 2 \end{pmatrix} = \begin{pmatrix} 8 \\ y \end{pmatrix}$$

$$\begin{pmatrix} 2 + 2x \\ 5 \end{pmatrix} = \begin{pmatrix} 8 \\ y \end{pmatrix}$$

$$2 + 2x = 8 \rightarrow x = 3$$

$$y = 5$$

$$x = 3 \rightarrow \alpha = 3$$

$$y = 5 \rightarrow \beta = 5$$

#6 Aplique el método de Gauss - Jordan para determinar el conjunto solución del sistema: (4 Pts)

$$\begin{cases} x + y + z + w = 5 \\ 2x + 3y + 2z + 2w = 13 \\ 3x + 4y + 3z + 3w = 18 \end{cases}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 2 & 3 & 2 & 2 & 13 \\ 3 & 4 & 3 & 3 & 18 \end{array} \right)$$

$$\begin{array}{l} -2 \cdot F_1 + \widetilde{F}_2 \\ -3 \cdot F_1 + \widetilde{F}_3 \end{array} \left(\begin{array}{cccc|c} \boxed{1} & 1 & 1 & 1 & 5 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 3 \end{array} \right)$$

$$\begin{array}{l} -F_2 + \widetilde{F}_1 \\ -F_2 + \widetilde{F}_3 \end{array} \left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 2 \\ 0 & \boxed{1} & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$\begin{array}{c} x \quad y \quad z \quad w \quad s \\ \left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 2 \\ 0 & \boxed{1} & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{array}$$

$$x + z + w = 2 \rightarrow x = 2 - z - w$$

$$y = 3$$

$$y = 3$$

$$S = (2 - z - w, y, w, z)$$

#7 Considere matrices A , B y X cuadradas de orden 2×2 . Se sabe que $(A+B)^T$ es invertible y que satisfacen la siguiente igualdad:

$$XA^T = 2I - (BX^T)^T \quad (A+B)^T \cdot X \rightarrow X = \left[(A+B)^T \right]^{-1}$$

a) Utilice propiedades matriciales para verificar que $X = 2(A^T + B^T)^{-1}$.

$$X \cdot A^T = 2I - (B \cdot X^T)^T$$

$$X \cdot A^T = 2I - (X^T)^T \cdot B^T \quad (A \cdot B)^T = B^T \cdot A^T$$

$$X \cdot A^T = 2I - X \cdot B^T \quad (X^T)^T = X$$

$$\underline{X \cdot A^T} + \underline{X \cdot B^T} = 2I \quad XA + XB = X(A+B)$$

$$X(A^T + B^T) = 2I$$

$$(A+B)^T = A^T + B^T$$

$$X(A+B)^T = 2I$$

$$X = 2I \cdot \left[(A+B)^T \right]^{-1}$$

$$(A+B)^T = (A+B)^T$$

Es una identidad

$$X = 2(A^T + B^T)^{-1}$$

$$AI = A$$

$$kI = k$$

b) Usando el resultado de la parte a) y si se cumple que

$$A+B = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix},$$

calcule explícitamente la matriz X .

$$X = 2 (A^T + B^T)^{-1}$$

$$= 2 [(A+B)^T]^{-1}$$

$$A+B = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$$

$$(A+B)^T = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$$

$$[(A+B)^T]^{-1} = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array} \right)$$

$$-3 \cdot F1 + \widetilde{F}2 \left(\begin{array}{cc|cc} \textcircled{1} & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array} \right)$$

$$-1 \cdot \widetilde{F}2 \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & \textcircled{1} & 3 & -1 \end{array} \right)$$

$$-2 \cdot F2 + \widetilde{F}1 \left(\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & \textcircled{1} & 3 & -1 \end{array} \right)$$

$$[(A+B)^T]^{-1} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}$$

$$\boxed{2 [(A+B)^T]^{-1} = \begin{pmatrix} -10 & 4 \\ 6 & -2 \end{pmatrix}}$$

P1 - S1 - 2023

1. [5 puntos] Considere el sistema de ecuaciones

$$\begin{cases} x + 2y + 3z = 7 \\ x + z = y \\ x + 5y + 5z - 14 = 0 \end{cases} \rightarrow \begin{aligned} x + 2 - y &= 0 \\ x - y + z &= 0 \\ x + 5y + 5z &= 14 \end{aligned}$$

donde x, y, z son incógnitas. Utilice el método de Gauss-Jordan

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 1 & -1 & 1 & 0 \\ 1 & 5 & 5 & 14 \end{array} \right)$$

$$\begin{aligned} -F_1 + \widetilde{F}_2 \\ -F_1 + \widetilde{F}_3 \end{aligned} \left(\begin{array}{ccc|c} \boxed{1} & 2 & 3 & 7 \\ 0 & -3 & -2 & -7 \\ 0 & 3 & 2 & 7 \end{array} \right)$$

$$-\frac{1}{3} \cdot \widetilde{F}_2 \left(\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & \boxed{1} & \frac{2}{3} & \frac{7}{3} \\ 0 & 3 & 2 & 7 \end{array} \right)$$

$$\begin{aligned} -2 \cdot F_2 + \widetilde{F}_1 \\ -3 \cdot F_2 + \widetilde{F}_3 \end{aligned} \left(\begin{array}{ccc|c} x & y & z & s \\ 1 & 0 & \frac{5}{3} & \frac{7}{3} \\ 0 & \boxed{1} & \frac{2}{3} & \frac{7}{3} \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{matrix} x \\ y \\ z \end{matrix}$$

$$x + \frac{5}{3}z = \frac{7}{3} \rightarrow x = \frac{7}{3} - \frac{5}{3}z$$

$$y + \frac{2}{3}z = \frac{7}{3} \rightarrow y = \frac{7}{3} - \frac{2}{3}z$$

$$S = \left(\frac{7}{3} - \frac{5}{3}z, \frac{7}{3} - \frac{2}{3}z, z \right), z \in \mathbb{R}$$

2. [4 puntos] Considere las matrices $A = \begin{pmatrix} 3 & 0 & 2 \\ 2 & -1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -3 \\ -4 & 0 \\ 2 & -1 \end{pmatrix}$ y $D = \begin{pmatrix} 6 & 1 \\ 3 & 1 \\ 3 & 0 \end{pmatrix}$.

Determine la matriz B tal que $A^T \cdot B - C = D$.

$$A^T = \begin{pmatrix} 3 & 2 \\ 0 & -1 \\ 2 & 1 \end{pmatrix} \quad A^T B = D + C$$

3×2

3×2 $A_C = B_F$

$$A^T \cdot B = \begin{pmatrix} 3 & 2 \\ 0 & -1 \\ 2 & 1 \end{pmatrix} \times \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{pmatrix} 3a+2c & 3b+2d \\ 0a-c & 0b-d \\ 2a+c & 2b+d \end{pmatrix}$$

$$D+C = \begin{pmatrix} 6 & 1 \\ 3 & 1 \\ 3 & 0 \end{pmatrix} + \begin{pmatrix} 2 & -3 \\ -4 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 8 & -2 \\ -1 & 1 \\ 5 & -1 \end{pmatrix}$$

$$A^T B = D+C$$

$$\begin{pmatrix} 3a+2c & 3b+2d \\ 0a-c & 0b-d \\ 2a+c & 2b+d \end{pmatrix} = \begin{pmatrix} 8 & -2 \\ -1 & 1 \\ 5 & -1 \end{pmatrix}$$

$$A^T B = D + C$$

$$\begin{pmatrix} 3a+2c & 3b+2d \\ 0a-c & 0b-d \\ 2a+c & 2b+d \end{pmatrix} = \begin{pmatrix} 8 & -2 \\ -1 & 1 \\ 5 & -1 \end{pmatrix}$$

$$-c = -1 \rightarrow c = 1$$

$$-d = 1 \rightarrow d = -1$$

$$2a+c=5 \quad 2b+d=-1$$

$$2a+1=5 \quad 2b-1=-1$$

$$2a=4 \quad 2b=0$$

$$a=2 \quad b=0$$

$$b = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \boxed{\begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}}$$

5. [5 puntos] Determine $z \in \mathbb{C}$ que satisfice simultáneamente las siguientes condiciones :

- $|\bar{z} + (1 - i)| = 5$

- $\text{Arg}(z - (1 + 2i)) = \frac{3\pi}{4}$

$$\frac{b}{a} = \tan(\theta)$$

$$\text{Arg}(z - (1 + 2i)) = \frac{3\pi}{4}$$

$$a + bi - 1 - 2i = \frac{3\pi}{4}$$

$$(a - 1) + (b - 2)i = \frac{3\pi}{4}$$

$$\frac{3\pi}{4}, \frac{180}{\pi} = 135$$

$$\frac{b - 2}{a - 1} = \tan\left(\frac{3\pi}{4}\right)$$

$$90 < 135 < 180$$

$$a < 0 \quad b > 0$$

$$\frac{b - 2}{a - 1} = -1$$

$$b - 2 = 1 - a$$

$$1 - a = b - 2$$

$$b = 1 + 2 - a$$

$$-a = b - 2 - 1$$

$$b = 3 - a$$

$$a = 3 - b$$

$$|\bar{z} + (1 - i)| = 5$$

$$|a - bi + 1 - i| = 5$$

$$|(a + 1) + (-b - 1)i| = 5$$

$$(a + 1)^2 + (-b - 1)^2 = 5^2 \quad a = 3 - b$$

$$(3 - b + 1)^2 + (-b - 1)^2 = 25$$

$$(4 - b)^2 + b^2 + 2b + 1 - 25 = 0 \quad (-a - b)^2$$

$$16 - 8b + b^2 + b^2 + 2b + 1 - 25 = 0 \quad (a + b)^2$$

$$2b^2 - 6b - 8 = 0$$

$$2b \quad 2 = 2b$$

$$b \quad -8 = -8b$$

$$-6b$$

$$(2b + 2)(b - 4)$$

$$(2b+2)(b-7)$$

$$2b+2=0$$

$$b=-1$$

$$b-7=0$$

$$b=7$$

$$b>0$$

$$a<0$$

$$a=3-b$$

$$=3-7$$

$$=-4$$

$$z = -1 + 7i$$

6. [4 puntos] Factorice en $\mathbb{C}$ el polinomio $P(z) = z^4 - 90z + 35z^2 - 10z^3 + 234$ si se sabe que $z = 5 - i$ es un cero de $P(z)$.

$$z^4 - 10z^3 + 35z^2 - 90z + 234$$

$$\begin{array}{r|rrrrr} 1 & -10 & 35 & -90 & 234 & \\ & 5-i & -26 & 45-9i & -234 & 5-i \\ \hline 1 & -5-i & 9 & -45-9i & 0 & (x-(5-i)) \end{array}$$

$$\begin{array}{r|rrrr} 1 & -5-i & 9 & -45-9i & \\ & 5+i & 0 & 95+9 & 5+i \\ \hline 1 & 0 & 9 & 0 & (x-(5+i)) \end{array}$$

$(5-i)(-5-i)$
 $-25 - 5i + 5i + i^2$
 $-25 - 1$
 -26

$$z^2 + 9 \quad a^2 - b^2 = (a-b)(a+b) \quad (5-i)(-5-i)$$

$$a^2 + b^2 = (a-bi)(a+bi) \quad -225 - 95i + 95i + 9i^2$$

$$-225 - 9$$

$$-234$$

$(z-3i)(z+3i)$

$$\begin{array}{ll} x-3i=0 & x+3i=0 \\ x=3i & x=-3i \end{array}$$

$$(x-(5-i))(x-(5+i))(x-3i)(x+3i)$$